IGCSE 0607 Practice Questions — E10.8 — Combined Questions & Markscheme

Cumulative frequency diagrams

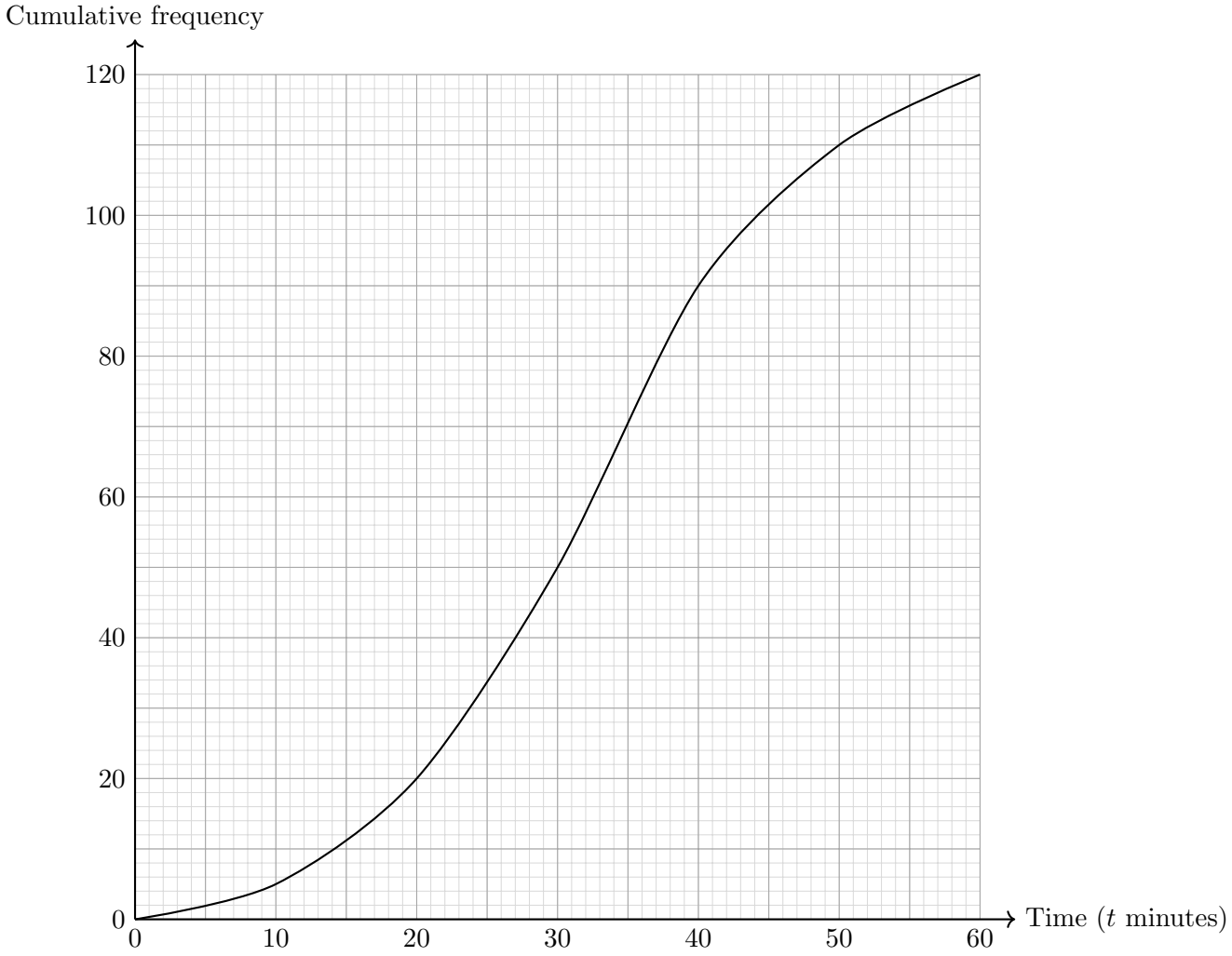
Total Questions: 17 — **Total Marks:** 127 Marks

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- **E10.8** Cumulative frequency diagrams
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(13 Qs)*

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1 The cumulative frequency diagram shows the times taken, t minutes, by 120 students to complete a test.



(a) Use the diagram to estimate the median time.

..... minutes **[1]**

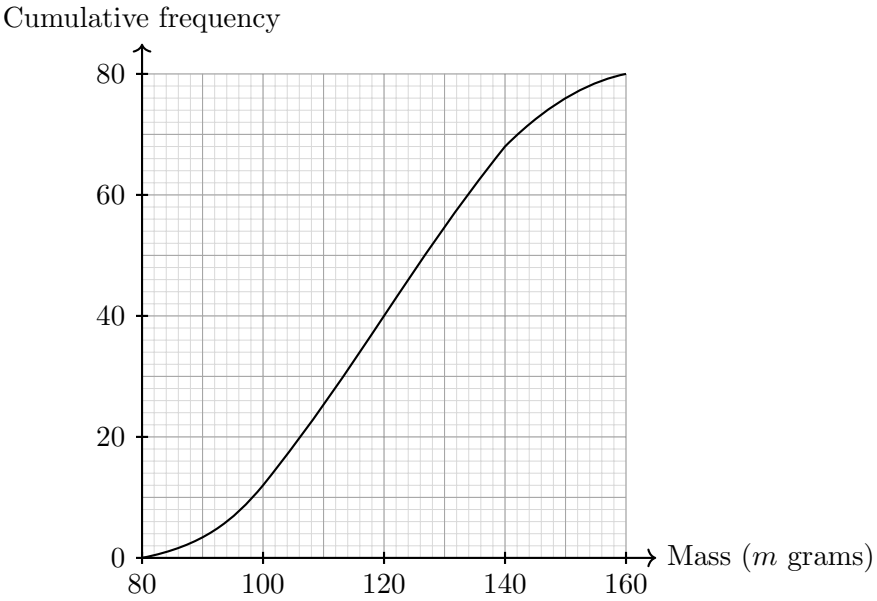
(b) Use the diagram to estimate the interquartile range.

..... minutes **[2]**

(c) Students who take longer than 45 minutes are given a penalty.
Estimate the number of students who are given a penalty.

..... **[2]**

2 The cumulative frequency diagram shows information about the masses, m grams, of 80 apples.



(a) Use the cumulative frequency diagram to find an estimate for

(i) the median,

..... g **[1]**

(ii) the lower quartile,

..... g **[1]**

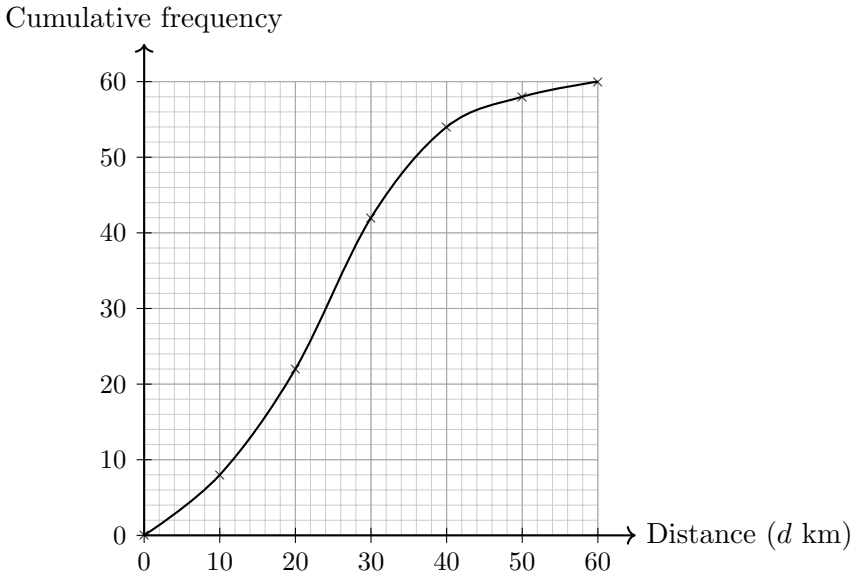
(iii) the interquartile range.

..... g **[2]**

(b) Find an estimate for the number of apples with a mass greater than 140 g.

..... **[1]**

3 The cumulative frequency diagram shows information about the distance, d km, travelled by 60 people to work.



(a) Use the diagram to find an estimate for:

(i) the median,

..... km **[1]**

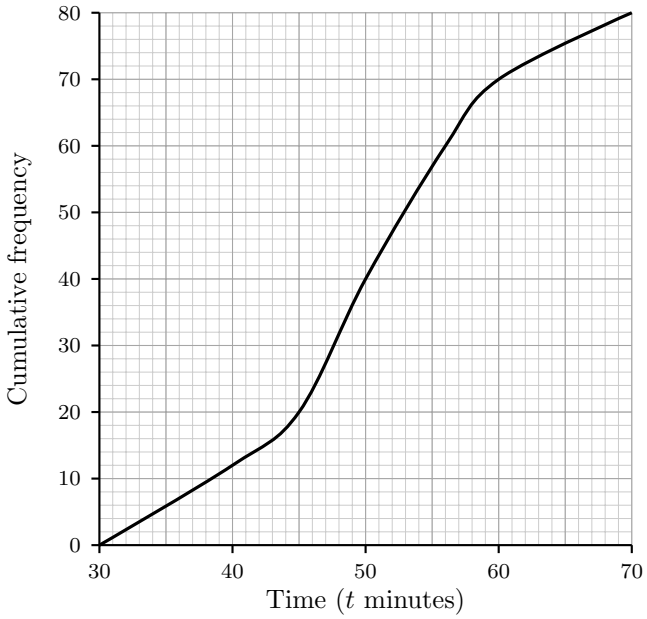
(ii) the interquartile range.

..... km **[2]**

(b) Find the number of people who travelled more than 45 km to work.

..... **[3]**

4 The cumulative frequency diagram shows information about the times, t minutes, taken by 80 runners to complete a 10 km race.



(a) Use the cumulative frequency diagram to find

(i) the median time,

..... minutes **[1]**

(ii) the lower quartile and the upper quartile,

Lower quartile = ... , Upper quartile = ... **[2]**

(iii) the interquartile range.

..... minutes **[1]**

(b) Find the number of runners who took more than 60 minutes to complete the race.

..... **[2]**

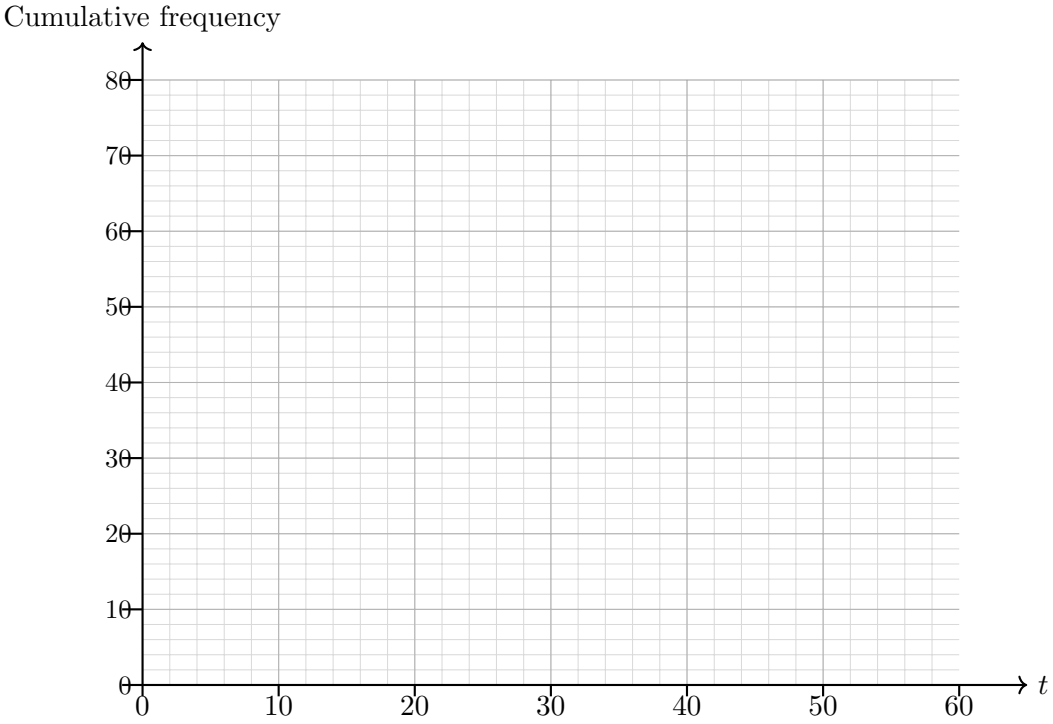
(c) Runners who finished in the fastest 15% of times qualified for a regional event.
Estimate the maximum qualifying time.

..... minutes **[2]**

5 The table shows the cumulative frequencies for the times, t minutes, taken by 80 students to complete a puzzle.

Time (t minutes)	$t \leq 10$	$t \leq 20$	$t \leq 30$	$t \leq 40$	$t \leq 50$	$t \leq 60$
Cumulative frequency	6	22	48	68	78	80

(a) On the grid, draw a cumulative frequency diagram to show this information.

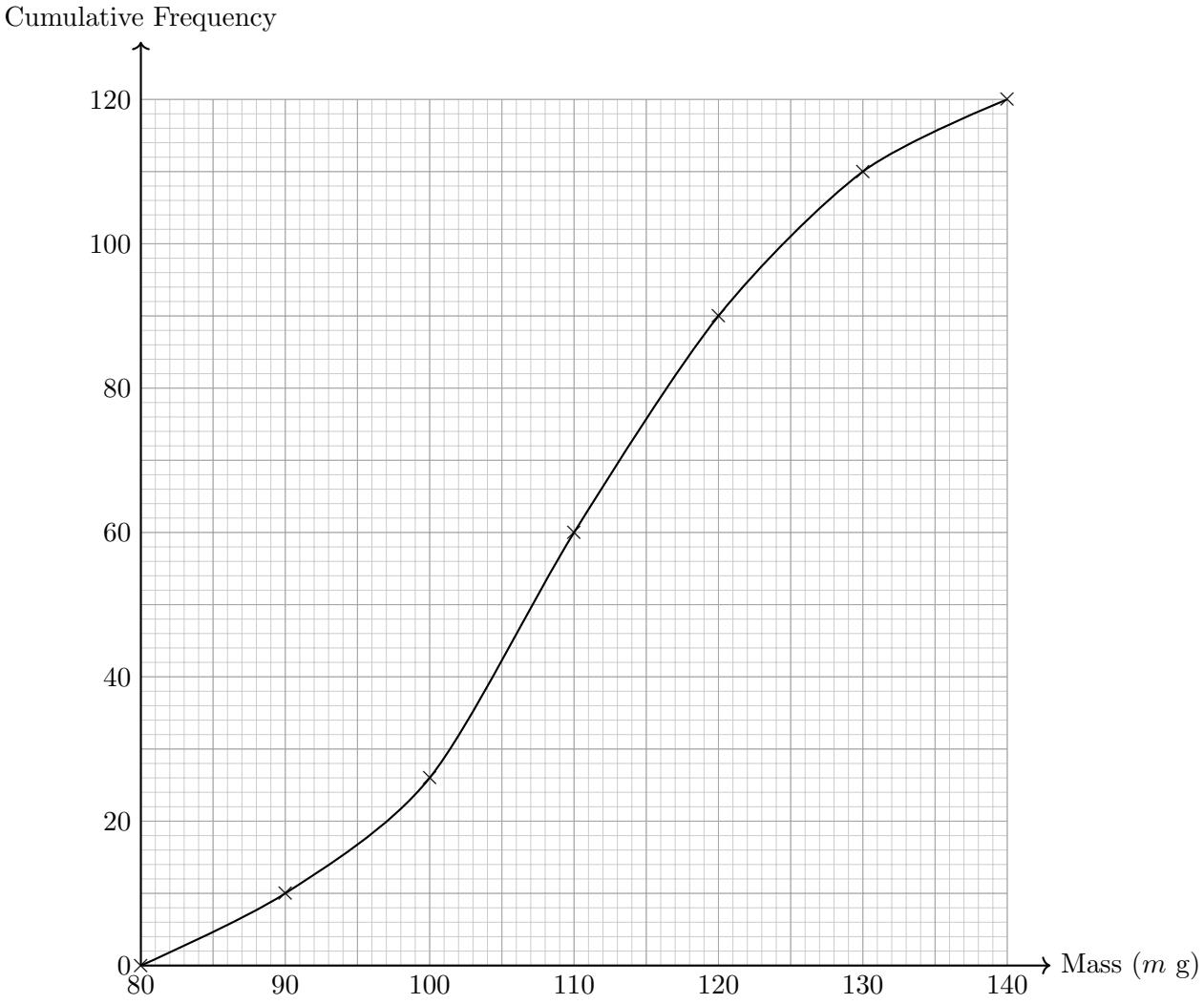


[3]

(b) Use your diagram to estimate the median time.

..... minutes **[2]**

6 The cumulative frequency diagram shows information about the masses, m grams, of 120 apples.



(a) Find the median mass.

..... g **[1]**

(b) Find the interquartile range.

..... g **[2]**

(c) Find the number of apples with a mass greater than 125 g.

..... **[2]**

7 The frequency table shows information about the times, t minutes, taken by 100 students to complete a puzzle.

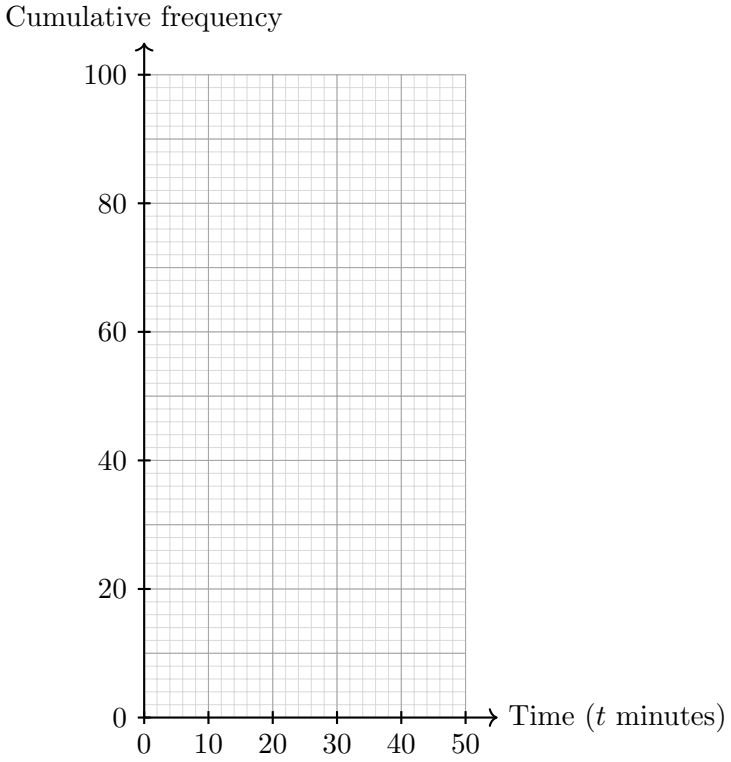
Time (t minutes)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$	$40 < t \leq 50$
Frequency	12	24	36	18	10

(a) Complete the cumulative frequency table.

Time (t minutes)	$t \leq 10$	$t \leq 20$	$t \leq 30$	$t \leq 40$	$t \leq 50$
Cumulative frequency	12				100

[1]

(b) On the grid, draw a cumulative frequency diagram to show this information.



[3]

(c) Use your diagram to find an estimate for the 80th percentile.

..... minutes **[1]**

(d) Students who take more than 35 minutes must take a second test.
Use your diagram to estimate the percentage of students who must take a second test.

..... % **[2]**

8 The grouped frequency table shows the time, t minutes, taken by 80 students to complete a science test.

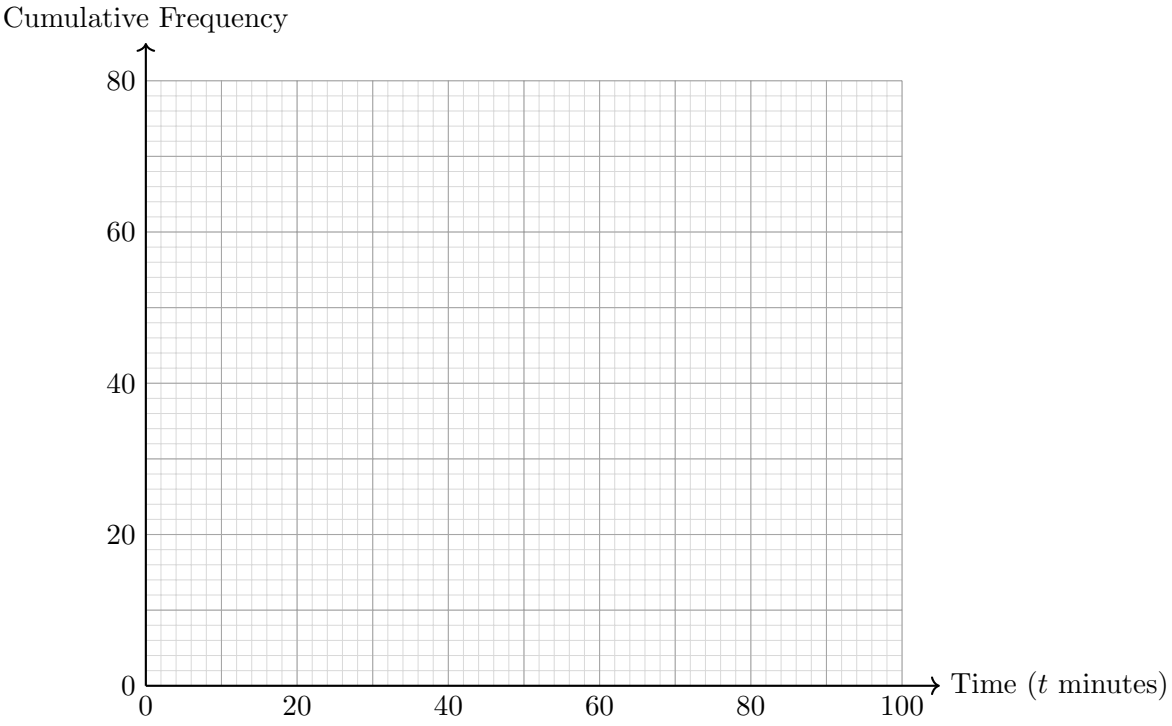
Time (t minutes)	$0 < t \leq 20$	$20 < t \leq 40$	$40 < t \leq 60$	$60 < t \leq 80$	$80 < t \leq 100$
Frequency	8	22	34	12	4

(a) Complete the cumulative frequency table.

Time (t minutes)	$t \leq 20$	$t \leq 40$	$t \leq 60$	$t \leq 80$	$t \leq 100$
Cumulative freq.	8				

[2]

(b) On the grid, draw a cumulative frequency diagram to show this information.



[3]

(c) Use your graph to estimate the number of students who took more than 55 minutes to complete the test.

..... **[2]**

9 A company records the salaries, s thousand dollars, of its 200 employees.
The frequency table shows the results.

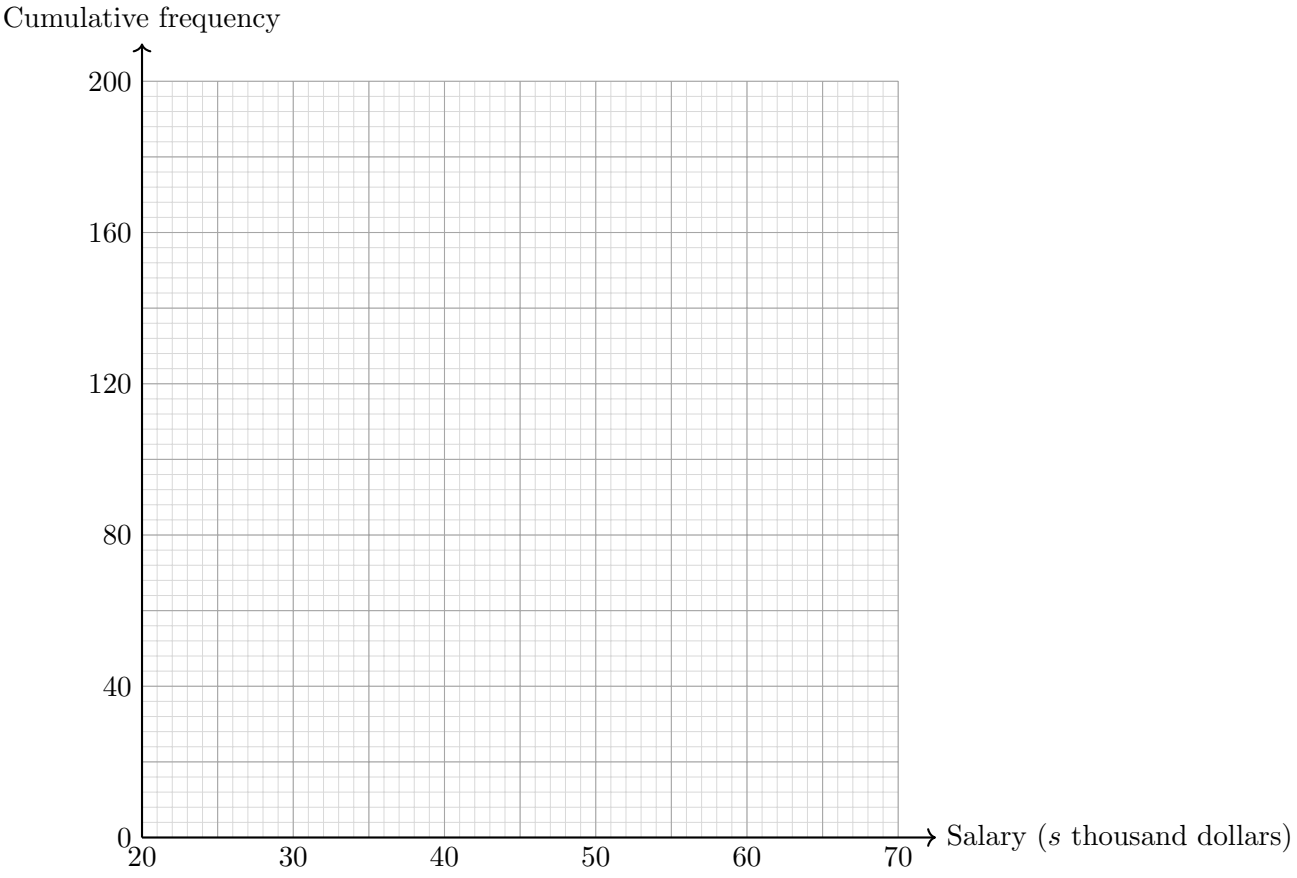
Salary (s thousand dollars)	$20 < s \leq 30$	$30 < s \leq 40$	$40 < s \leq 50$	$50 < s \leq 60$	$60 < s \leq 70$
Frequency	24	66	70	28	12

(a) Complete the cumulative frequency table.

Salary (s thousand dollars)	$s \leq 30$	$s \leq 40$	$s \leq 50$	$s \leq 60$	$s \leq 70$
Cumulative frequency	24				

[2]

(b) On the grid, draw a cumulative frequency diagram to show this information.

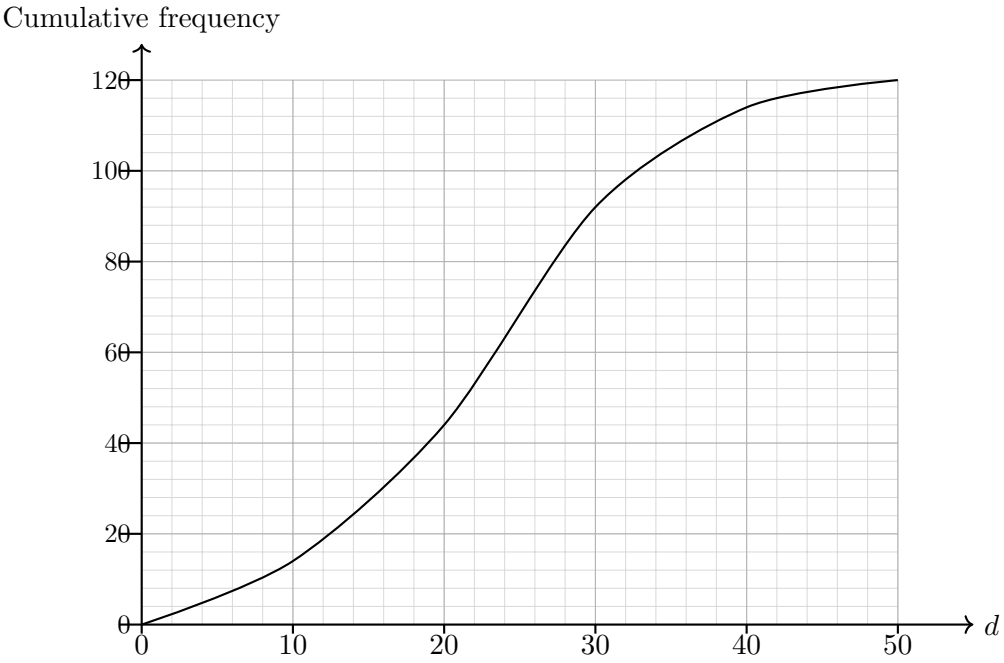


[3]

(c) Use your diagram to estimate the 70th percentile of the salaries.

..... thousand **[2]**

10 The cumulative frequency diagram shows information about the distances, d km, travelled by 120 people to work.



(a) Use the diagram to estimate the median distance.

..... km **[2]**

(b) Use the diagram to estimate the interquartile range.

..... km **[3]**

(c) Estimate the number of people who travelled more than 35 km to work.

..... **[3]**

11 A table shows information about the masses of 120 apples.

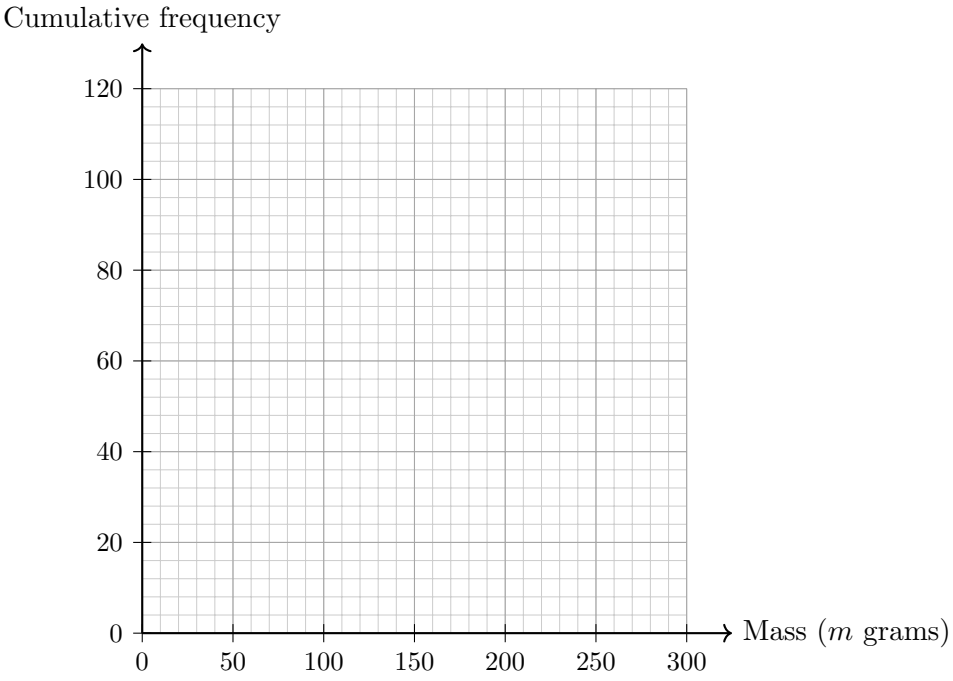
Mass (m grams)	$50 < m \leq 100$	$100 < m \leq 150$	$150 < m \leq 200$	$200 < m \leq 250$	$250 < m \leq 300$
Frequency	12	34	46	20	8

(a) Complete the cumulative frequency table.

Mass (m grams)	$m \leq 100$	$m \leq 150$	$m \leq 200$	$m \leq 250$	$m \leq 300$
Cumulative frequency	12				120

[2]

(b) On the grid, draw a cumulative frequency diagram to show this information.



[3]

(c) Use your diagram to estimate the 60th percentile.

..... g **[2]**

(d) Apples with a mass greater than 220 grams are classified as large.
Estimate the number of large apples.

..... **[3]**

12 A supermarket records the checkout waiting time, w seconds, for 120 customers.
The grouped frequency table shows the results.

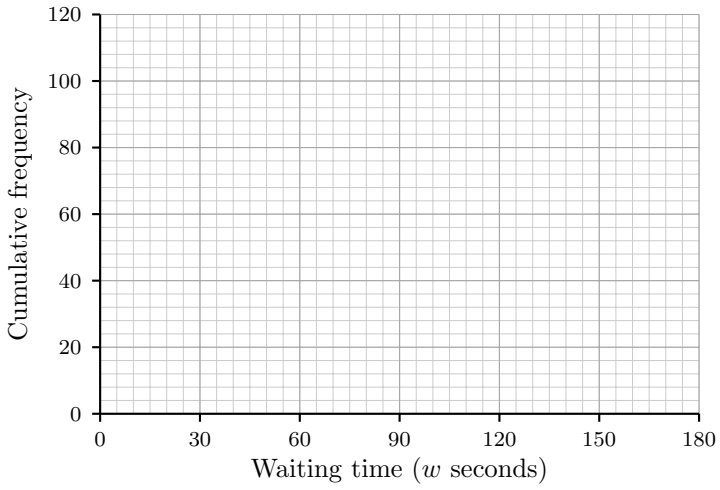
Waiting time (w s)	Frequency
$0 < w \leq 30$	10
$30 < w \leq 60$	22
$60 < w \leq 90$	44
$90 < w \leq 120$	28
$120 < w \leq 150$	12
$150 < w \leq 180$	4

(a) Complete the cumulative frequency table below.

Waiting time (w s)	Cumulative frequency
$w \leq 30$	10
$w \leq 60$	
$w \leq 90$	
$w \leq 120$	
$w \leq 150$	
$w \leq 180$	

[2]

(b) On the grid below, draw a cumulative frequency diagram to represent this information.
Mark your points clearly with small crosses ($\times$) and join them with a smooth curve.



[3]

(c) Use your cumulative frequency diagram to estimate

(i) the median waiting time,

..... seconds **[1]**

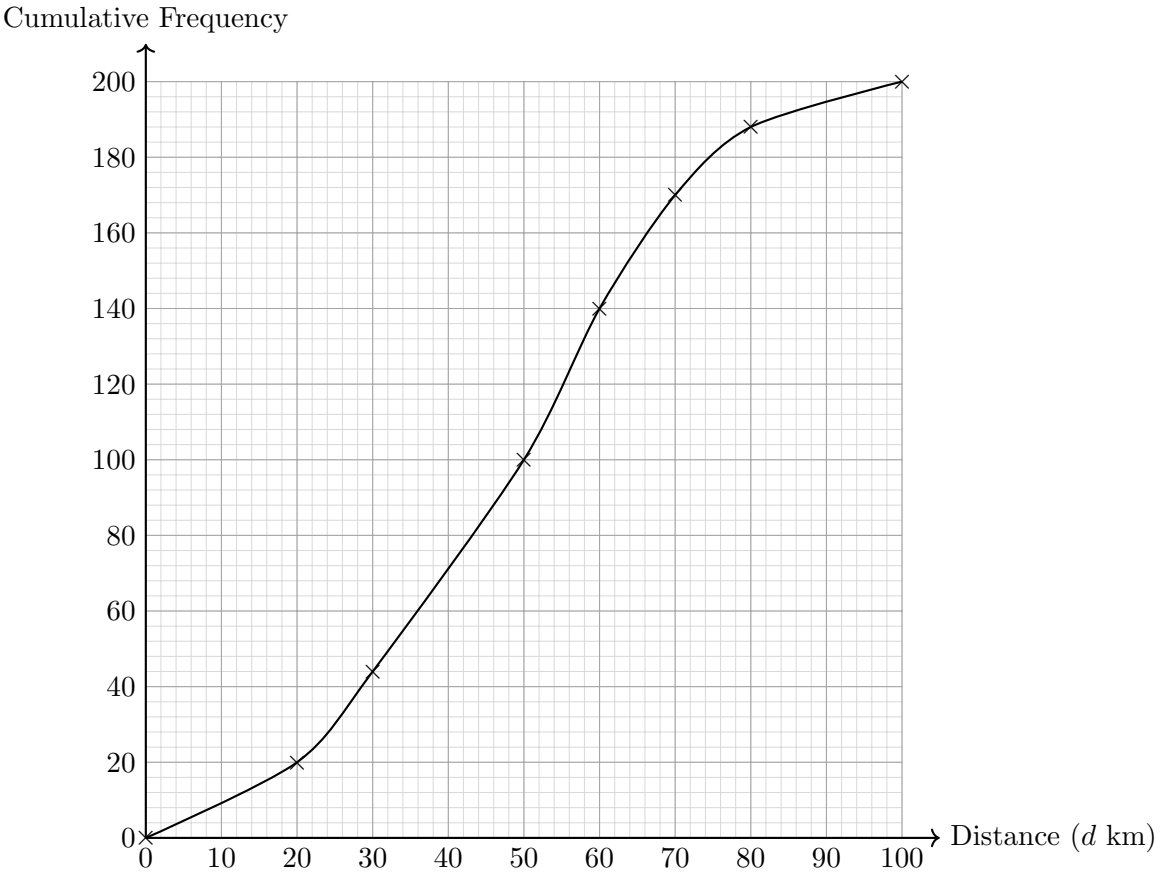
(ii) the 80th percentile.

..... seconds **[2]**

(d) The supermarket manager sets a target that at least 85% of customers must wait no more than 115 seconds.
Use your diagram to determine whether this target is met, showing your working clearly.

..... **[3]**

13 The cumulative frequency diagram shows information about the distance, d kilometres, travelled by 200 delivery vans in a single day.



(a) Find an estimate for the 60th percentile.

..... km **[2]**

(b) 15% of the vans travelled further than k km.
Find the value of k.

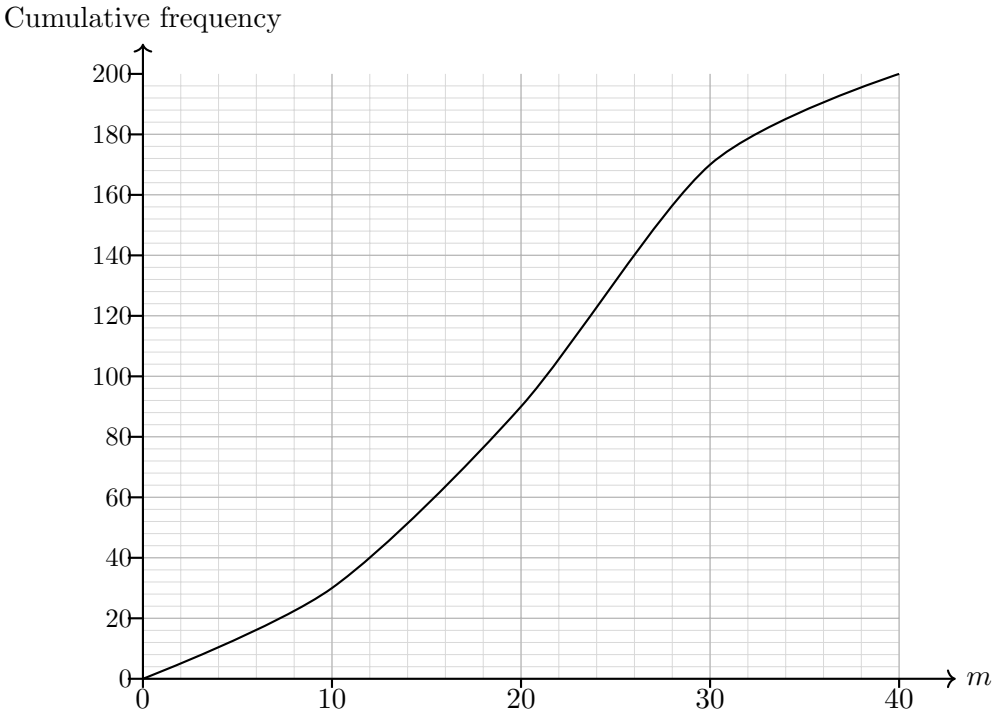
$k =$ ..... **[2]**

(c) Use the cumulative frequency diagram to complete the frequency table.

Distance (d km)	$0 < d \leq 30$	$30 < d \leq 50$	$50 < d \leq 80$	$80 < d \leq 100$
Frequency				

[3]

14 The cumulative frequency diagram shows information about the masses, m grams, of 200 parcels.



(a) Find an estimate for the 80th percentile.

..... grams **[2]**

(b) Use the cumulative frequency diagram to complete the frequency table.

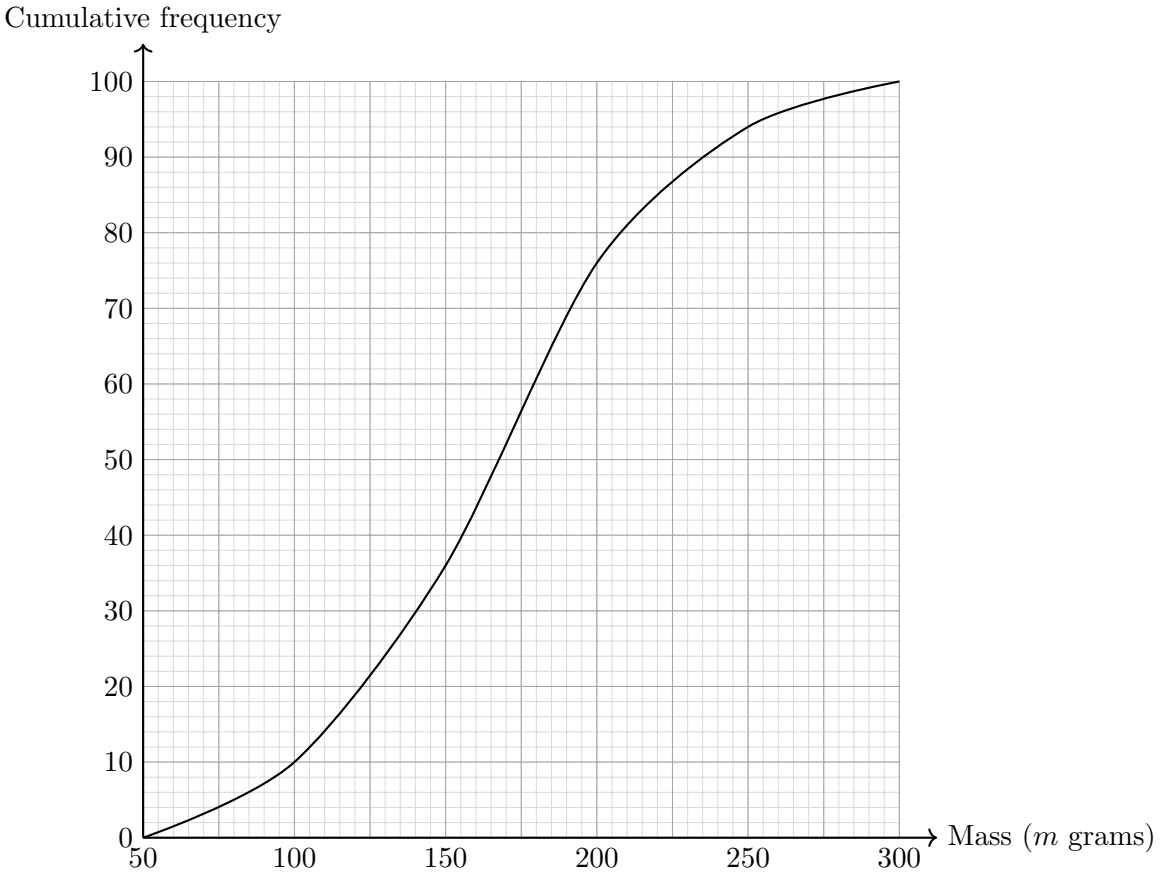
Mass (m grams)	$0 < m \leq 10$	$10 < m \leq 20$	$20 < m \leq 30$	$30 < m \leq 40$
Frequency				

[2]

(c) Calculate an estimate of the mean mass.

..... grams [4]

15 The cumulative frequency curve shows information about the masses, m grams, of 100 potatoes.



(a) Use the curve to estimate the number of potatoes with a mass greater than 220 grams.

..... **[3]**

(b) The curve was drawn using grouped data.
Find the values of p, q, and r.

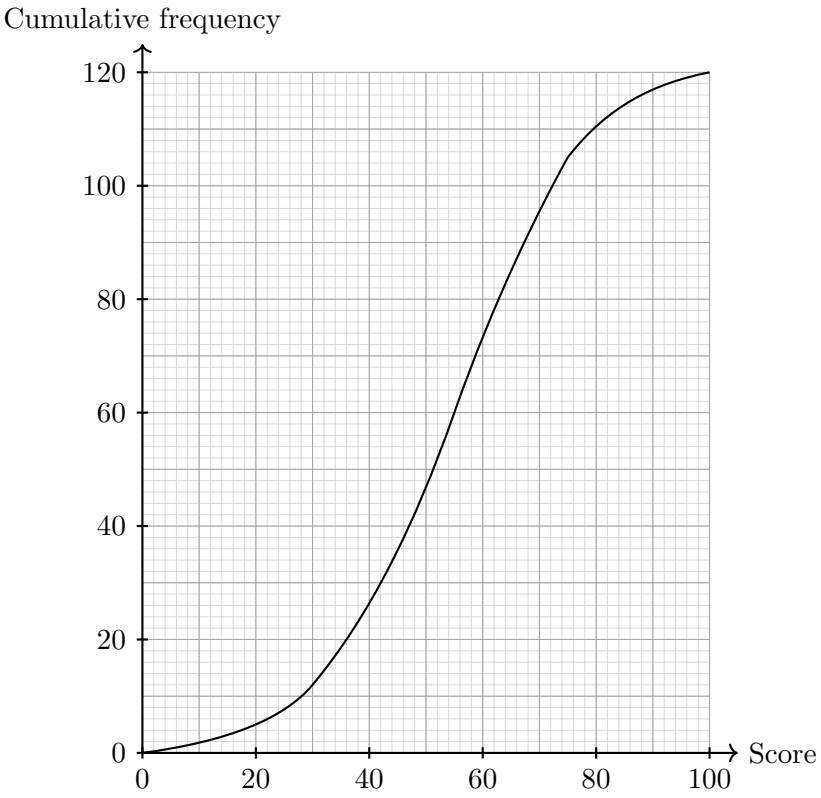
Mass (m grams)	$50 < m \leq 100$	$100 < m \leq 150$	$150 < m \leq 200$	$200 < m \leq 250$	$250 < m \leq 300$
Frequency	p	q	40	r	6

$p =$ ..... , $q =$ ..... , $r =$ ..... **[3]**

(c) Calculate an estimate of the mean mass of the potatoes.

..... g **[3]**

16 The cumulative frequency diagram shows information about the test scores of 120 students in Mathematics.
The maximum possible score is 100.



(a) Find an estimate for the interquartile range of the Mathematics scores.

..... **[2]**

(b) The top 15% of students are awarded a distinction.
Use the diagram to estimate the minimum score needed for a distinction.

..... **[2]**

(c) The same 120 students also took a Physics test.
For the Physics scores, the median is 62 and the interquartile range is 16.
Make two comparisons between the Mathematics scores and the Physics scores.

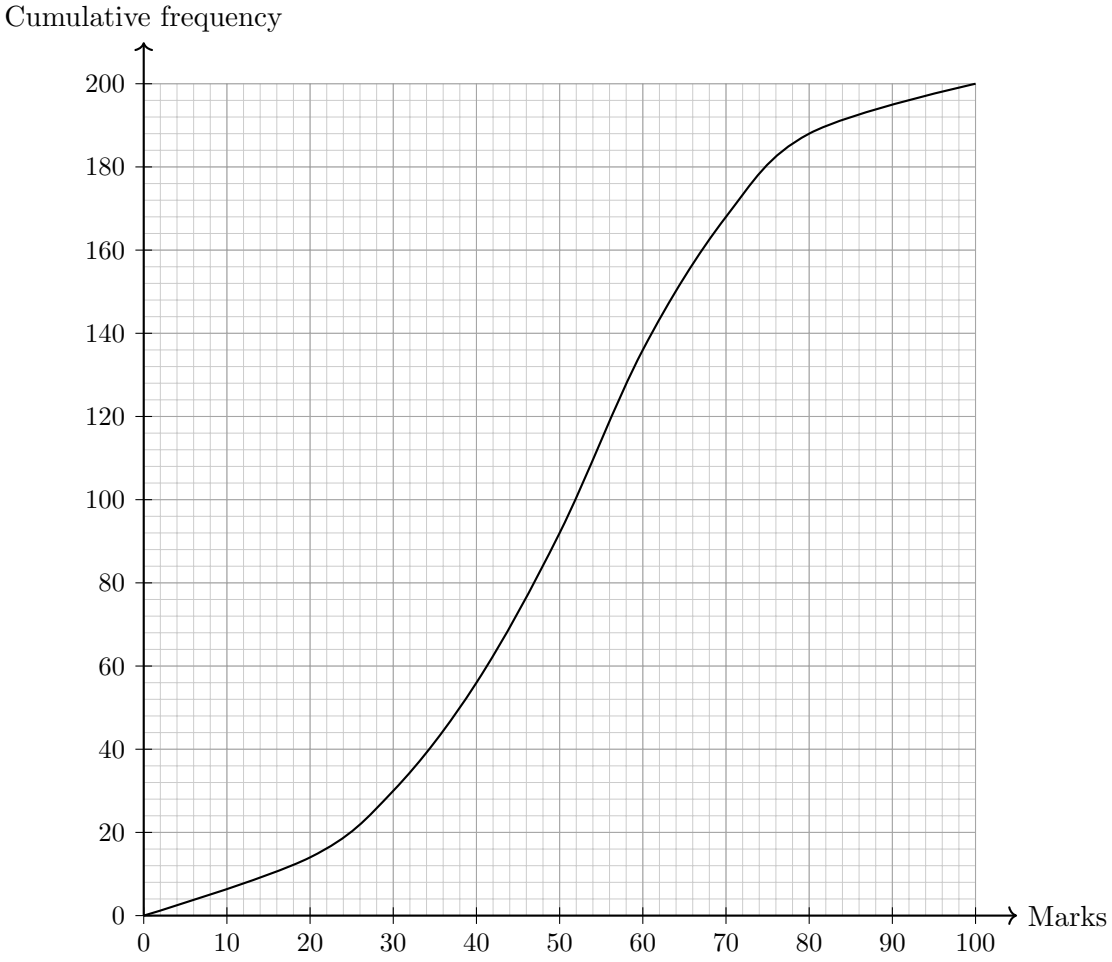
.....

..... **[2]**

(d) Two of the 120 students are chosen at random.
Using the cumulative frequency diagram, calculate an estimate for the probability that both students scored more than 60 in Mathematics.

..... **[3]**

17 The cumulative frequency diagram shows the distribution of marks scored by 200 candidates in an examination.



(a) Use your graph to estimate the interquartile range.

..... **[2]**

(b) 15% of the candidates score less than m marks.
Find the value of m.

$m =$ ..... **[2]**

(c) Candidates scoring more than P marks are awarded a Distinction.
Given that 30 candidates are awarded a Distinction, find the value of P.

$P =$ ..... **[2]**

(d) Candidates who score less than 45 marks have to take a supplementary test.
Calculate the percentage of candidates who do NOT have to take this test.

..... % **[4]**

1 **(a)** 32 to 34 **[B1]**

(b) Reading Upper Quartile as 40 and Lower Quartile as 23 to 24 **[M1]**

16 to 17.5 **[A1]**

(c) 120 − 100 (or 120 − their reading at $t = 45$) **[M1]**

18 to 22 **[A1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** **(i)** 120 **[B1]**

(ii) 108 (allow 107 to 109) **[B1]**

(iii) Upper quartile = 132 (allow 131 to 133) seen **[M1]**

24 (allow 22 to 26, or their UQ − their LQ) **[A1]**

(b) 80 − 68 = 12 **[B1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** **(i)** 24 **[B1]**

(ii) Reading Lower Quartile = 15 and Upper Quartile = 32.5 (Accept 32 to 33) **[M1]**

17.5 (Accept 17 to 18) **[A1]**

(b) Reading Cumulative Frequency at 45 km = 56 (Accept 55 to 57) **[M1]**

60 − 56 **[M1]**

4 (Accept 3 to 5) **[A1]**

[Back to Question] *[Total Marks: 6]*

4 **(a)** **(i)** 50 **[B1]**

(ii) Lower quartile = 45 **[B1]**

Upper quartile = 56 **[B1]**

(iii) 56 − 45 = 11 **[B1]**

(b) Reading cumulative frequency at $t = 60$ gives 70 **[M1]**

80 − 70 = 10 **[A1]**

(c) 15% of 80 = 12 runners **[M1]**

Reading time at cumulative frequency 12 gives 40 **[A1]**

[Back to Question] *[Total Marks: 8]*

5 **(a)** 6 points plotted correctly at the upper boundaries **[P2]**

(Allow P1 for 4 or 5 points plotted correctly)

A smooth curve drawn passing through all their points **[C1]**

(b) Reading from cumulative frequency 40 **[M1]**

$26 \leq$ answer ≤ 28 **[A1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** 110 **[B1]**

(b) Upper quartile = 120 AND Lower quartile = 101 to 102 (reads CF at 30) **[M1]**

18 to 19 **[A1]**

(c) Reads cumulative frequency at 125 g as 99 to 101 **[M1]**

19 to 21 (from $120 - 100$) **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** 36, 72, 90 correctly entered in the table **[B1]**

(b) Plots $(10, 12)$, $(20, 36)$, $(30, 72)$, $(40, 90)$, $(50, 100)$ accurately **[P2]**

Allow [P1] if at least 3 points are plotted accurately, or if all 5 points are plotted at class midpoints

Smooth increasing curve drawn through all plotted points starting at $(0, 0)$ **[C1]**

(c) 34 (allow 33 to 35, or correct reading from their curve at cumulative frequency 80) **[B1]**

(d) Reads cumulative frequency at $t = 35$ as 82 (allow 80 to 84) **[M1]**

18 (allow 16 to 20, or $100 -$ their reading) **[A1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** Values: $30, 64, 76, 80$ **[B2]**

Allow [B1] for 2 or 3 correct values.

(b) Points plotted correctly horizontally at upper class boundaries $(20, 40, 60, 80, 100)$ **[B1]**

Points plotted correctly vertically at heights $8, 30, 64, 76, 80$ **[B1]**

Smooth curve drawn through the points, anchored at $(0, 0)$ **[B1]**

(c) Reads curve at $t = 55$ correctly (expected value 53 to 57) **[M1]**

23 to 27 (Calculation: $80 -$ their CF value) **[A1]**

[Back to Question] *[Total Marks: 7]*

9 **(a)** $90, 160, 188, 200$ **[B2]**

(B1 for 2 or 3 correct values)

(b) 5 points plotted correctly from their table **[B2]**

(B1 for 3 or 4 points plotted correctly)

Smooth curve drawn through their points **[B1]**

(c) Reading from curve at cumulative frequency = 140 **[M1]**

46 to 48 **[A1]**

[Back to Question] *[Total Marks: 7]*

10 **(a)** Reading from cumulative frequency 60 **[M1]**

$22 \leq$ answer ≤ 24 **[A1]**

(b) Attempt to read from cumulative frequencies 90 (UQ) and 30 (LQ) **[M1]**

Lower quartile ≈ 15.5 and Upper quartile ≈ 29.5 seen **[M1]**

$12 \leq$ answer ≤ 16 **[A1]**

(c) Attempt to read at $d = 35$ (approx 105) **[M1]**

120 − their reading **[M1]**

$13 \leq$ answer ≤ 17 **[A1]**

[Back to Question] *[Total Marks: 8]*

11 **(a)** 46, 92, 112 **[B2]**

(Allow [B1] for 2 correct values)

(b) Points correctly positioned horizontally and vertically starting at $(50, 0)$ **[B1]**

All 5 points plotted correctly at class upper boundaries (Tolerance ± half small square) **[B1]**

Smooth curve drawn through all 6 points **[B1]**

(c) Finding target Cumulative Frequency = 60% of 120 = 72 **[M1]**

178 (Accept 176 to 180) **[A1]**

(d) Reading Cumulative Frequency at $m = 220$ is 100 **[M1]**

120 − 100 **[M1]**

20 (Accept 18 to 22) **[A1]**

[Back to Question] *[Total Marks: 10]*

12 **(a)** All five missing values correct: $32, 76, 104, 116, 120$ **[B2]**

Award [B1] if at least 3 of the 5 values are correct

(b) All 6 upper-boundary points $(30, 10), (60, 32), (90, 76), (120, 104), (150, 116), (180, 120)$ plotted accurately as small crosses **[B2]**

Smooth increasing curve drawn through all plotted points (extending from $(0, 0)$ or $(30, 10)$ to $(180, 120)$) **[B1]**

(c) **(i)** Answer in the range 78 to 82 (reading at cumulative frequency 60) **[B1]**

(ii) $0.80 \times 120 = 96$ **[M1]**

Answer in the range 108 to 113 (reading at cumulative frequency 96) **[A1]**

(d) Reading cumulative frequency at $w = 115$ gives approximately 100 customers (accept 98 to 101) **[M1]**

Percentage $= \frac{100}{120} \times 100 = 83.3\%$ (or 85% of 120 = 102 customers) **[M1]**

No, the target is not met because $83.3\% < 85\%$ (or $100 < 102$) **[A1]**

[Back to Question] *[Total Marks: 11]*

13 (a) Calculates 60% of 200 = 120 **[M1]**

54 to 56 **[A1]**

(b) Calculates 15% of 200 = 30, and targets CF of $200 - 30 = 170$ **[M1]**

70 **[A1]**

(c) Frequency for $0 < d \leq 30$ is 44 **[B1]**

Frequency for $30 < d \leq 50$ is 56 $(100 - 44)$ **[B1]**

Frequencies for $50 < d \leq 80$ is 88 AND $80 < d \leq 100$ is 12 **[B1]**

[Back to Question] *[Total Marks: 7]*

14 (a) $0.8 \times 200 = 160$ **[M1]**

$28 \leq$ answer ≤ 29 **[A1]**

(b) Frequencies 30, 60, 80, 30 correct **[B2]**

(Allow B1 for 2 or 3 correct frequencies)

(c) Using midpoints 5, 15, 25, 35 **[M1]**

$\Sigma fx = 30 \times 5 + 60 \times 15 + 80 \times 25 + 30 \times 35$ (using their frequencies) **[M1]**

their $4100 \div 200$ **[M1]**

20.5 **[A1]**

[Back to Question] *[Total Marks: 8]*

15 (a) Reading from curve at $m = 220$: 81 to 85 **[M1]**

100 − their reading **[M1]**

15 to 19 **[A1]**

(b) $p = 10$ **[B1]**

$q = 26$ **[B1]**

$r = 18$ **[B1]**

(c) Use of correct midpoints: $75, 125, 175, 225, 275$ **[M1]**

$\sum f \times m = 10 \times 75 + 26 \times 125 + 40 \times 175 + 18 \times 225 + 6 \times 275 = 16700$ **[M1]**

$16700 \div 100 = 167$ **[A1]**

[Back to Question] *[Total Marks: 9]*

16 **(a)** Upper quartile = 66 (allow 65 to 67) and Lower quartile = 42 (allow 41 to 43) **[M1]**

24 (allow 22 to 26) **[A1]**

(b) Identifies 85th percentile: $0.85 \times 120 = 102$ **[M1]**

73 (allow 72 to 74) **[A1]**

(c) On average, the Physics scores are higher because the Physics median (62) is greater than the Mathematics median (55) **[B1]**

The Physics scores are more consistent (less spread out) because the Physics IQR (16) is smaller than the Mathematics IQR (24) **[B1]**

(d) Reads cumulative frequency at score 60 as 72 (allow 71 to 73), giving $120 - 72 = 48$ students scoring more than 60 **[B1]**

$\frac{48}{120} \times \frac{47}{119}$ **[M1]**

$\frac{94}{595}$ or 0.158 (allow 0.151 to 0.165) **[A1]**

[Back to Question] *[Total Marks: 9]*

17 **(a)** Reading Lower Quartile = 38 (Accept 37 to 39) and Upper Quartile = 64 (Accept 63 to 66) **[M1]**

26 (Accept 24 to 29) **[A1]**

(b) Finding Cumulative Frequency = 15% of 200 = 30 **[M1]**

30 (Accept 29 to 31) **[A1]**

(c) Finding target Cumulative Frequency = $200 - 30 = 170$ **[M1]**

71 (Accept 70 to 72) **[A1]**

(d) Reading Cumulative Frequency at 45 marks = 74 (Accept 72 to 76) **[M1]**

Number of candidates NOT taking test = $200 - 74 = 126$ **[M1]**

$\frac{126}{200} \times 100\%$ **[M1]**

63 (Accept 62 to 64) **[A1]**

[Back to Question] *[Total Marks: 10]*